

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has a diameter with endpoints $(2, 4)$ and $(2, 14)$. An equation of this circle is $(x - 2)^2 + (y - 9)^2 = r^2$, where r is a positive constant. What is the value of r ?

Correct Answer: 5

Rationale

The correct answer is **5**. The standard form of an equation of a circle in the xy -plane is $(x - h)^2 + (y - k)^2 = r^2$, where h , k , and r are constants, the coordinates of the center of the circle are (h, k) , and the length of the radius of the circle is r . It's given that an equation of the circle is $(x - 2)^2 + (y - 9)^2 = r^2$. Therefore, the center of this circle is $(2, 9)$. It's given that the endpoints of a diameter of the circle are $(2, 4)$ and $(2, 14)$. The length of the radius is the distance from the center of the circle to an endpoint of a diameter of the circle, which can be found using the distance formula, $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$. Substituting the center of the circle $(2, 9)$ and one endpoint of the diameter $(2, 4)$ in this formula gives a distance of $\sqrt{(2 - 2)^2 + (9 - 4)^2}$, or $\sqrt{0^2 + 5^2}$, which is equivalent to **5**. Since the distance from the center of the circle to an endpoint of a diameter is **5**, the value of r is **5**.